

School of Sciences
(Department of Mathematics)

APPLIED CALCULUS LAB
(26SOSP103)

MATLAB LABORATORY MANUAL



B.Tech. Programme (first sem.)
Academic Session: 2026-27

Name:

Roll No:

Group:

Instructor:

S. No.	List of Experiments	Date	Signature
1	Introduction to MATLAB: Basic commands, symbolic variables, arithmetic operations, and plotting of standard functions		
2	Graphing functions and derivatives		
3	Successive differentiation and Leibnitz's Theorem of algebraic, trigonometric, exponential, and logarithmic functions		
4	Taylor and Maclaurin series expansion of one variable functions and approximation of function values		
5	Graphical verification of Rolle's Theorem and Lagrange's Mean Value Theorem		
6	Homogenous functions, Euler's theorem and Multivariable Taylor series		
7	Determination and Visualization of Maxima and Minima of Functions of Two Variables		
8	Gradient Descent for Function Minimization		
9	Newton's method for Function Minimization		
10	Single and Multiple Linear Regression using Gradient Descent		

